

EX. NO. 05

Use Autopsy to Create a Case and Import Evidence

Date: 12-08-2026

1. Aim

To create a forensic case in Autopsy, import a split E01 evidence image, process the evidence using ingest modules, examine the extracted artifacts, and generate a forensic report.

2. Software and Evidence

Item	Details
Forensic tool	Autopsy 4.21.0
Evidence type	Disk image
Image files	4Dell Latitude CPi.E01 and 4Dell Latitude CPi.E02
Data-source type	Disk Image or VM File
Time zone	Asia/Calcutta (GMT+05:30)
Case type	Single-user

The E01 and E02 files were kept in the same evidence directory. The E01 file was selected in Autopsy as the first segment of the image set.

3. Theory

Autopsy is a digital-forensics platform used to examine disk images and organize extracted artifacts. It can identify file types, deleted files, operating-system information, web activity, e-mail, metadata, and other evidence.

A case stores the investigation information, data-source details, ingest results, analysis findings, and generated reports.

4. Procedure

Step 1: Create a New Case

Autopsy was opened and New Case was selected. The following case information was entered:

text

Case Name: EX05_NIRANJANu

Case Number: 01

Examiner: NIRANJANU

Case Type: Single-User

The case was stored in the Ex. No. 5 case directory.

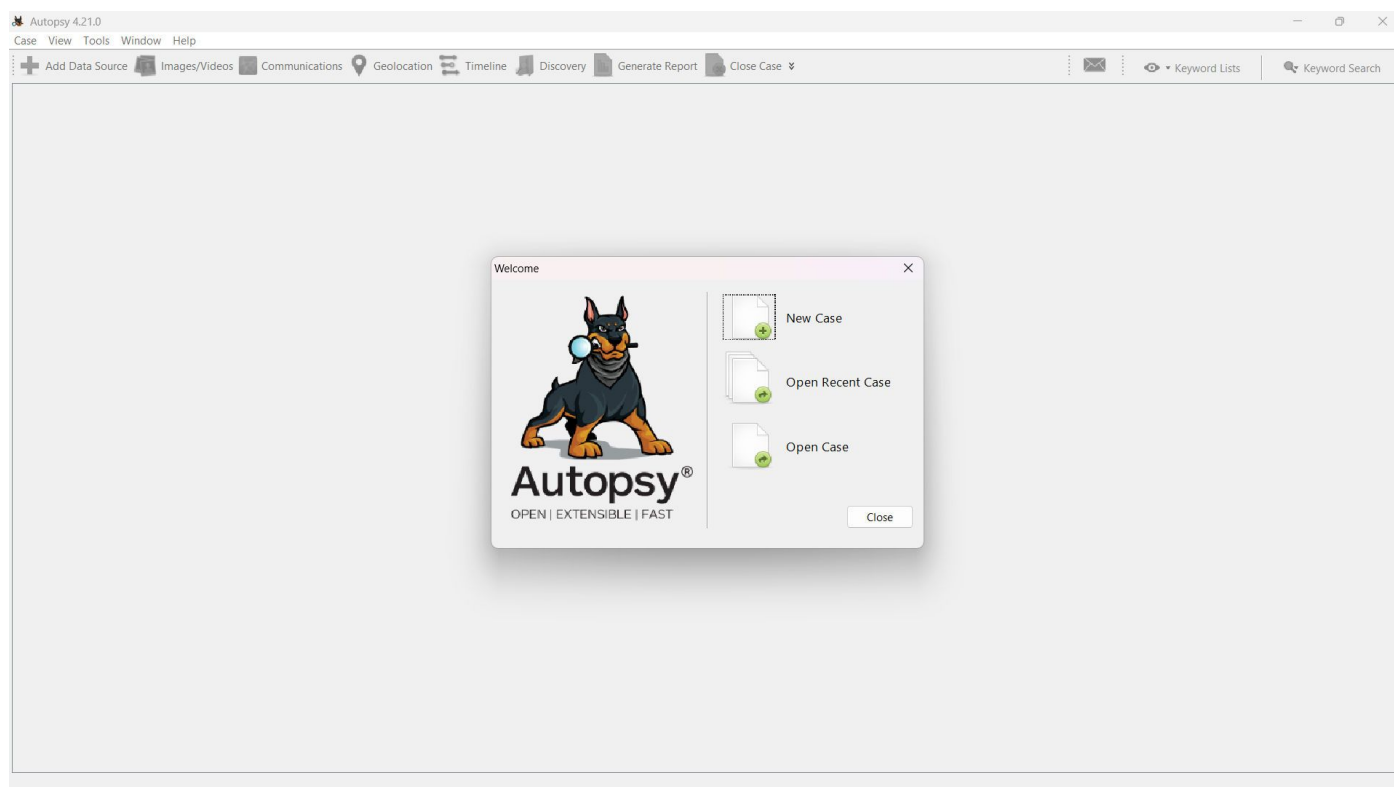


Figure 1: Autopsy Welcome window.

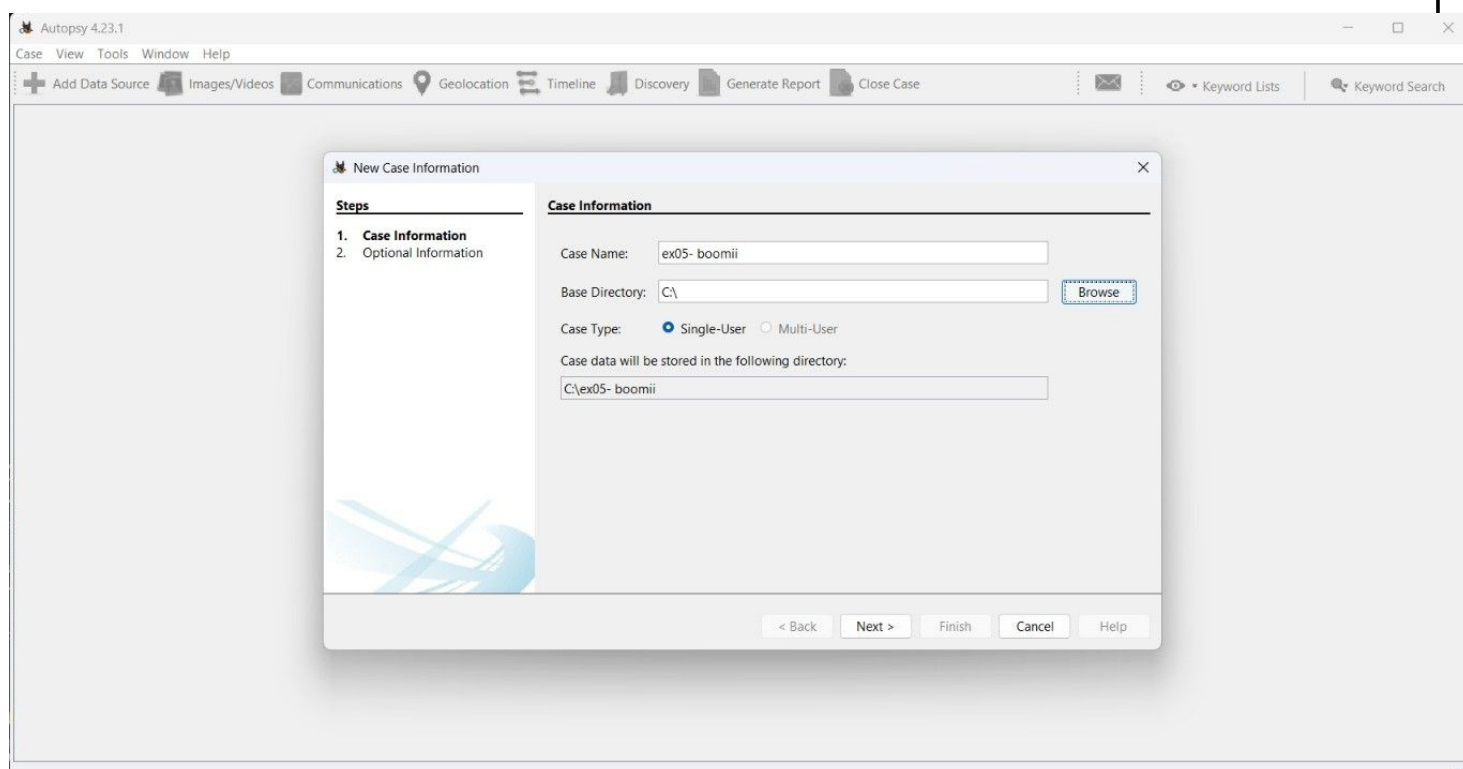


Figure 2: New Case Information window.

Step 2: Add the Evidence Source

The Add Data Source option was selected. The following source type was chosen:

text

Disk Image or VM File

The evidence path was selected as:

text

D:\n\df\Evidence\4Dell Latitude Cpi.E01

The time zone was set to:

text

(GMT+5:30) Asia/Calcutta

The associated 4Dell Latitude Cpi.E02 file was stored in the same directory.

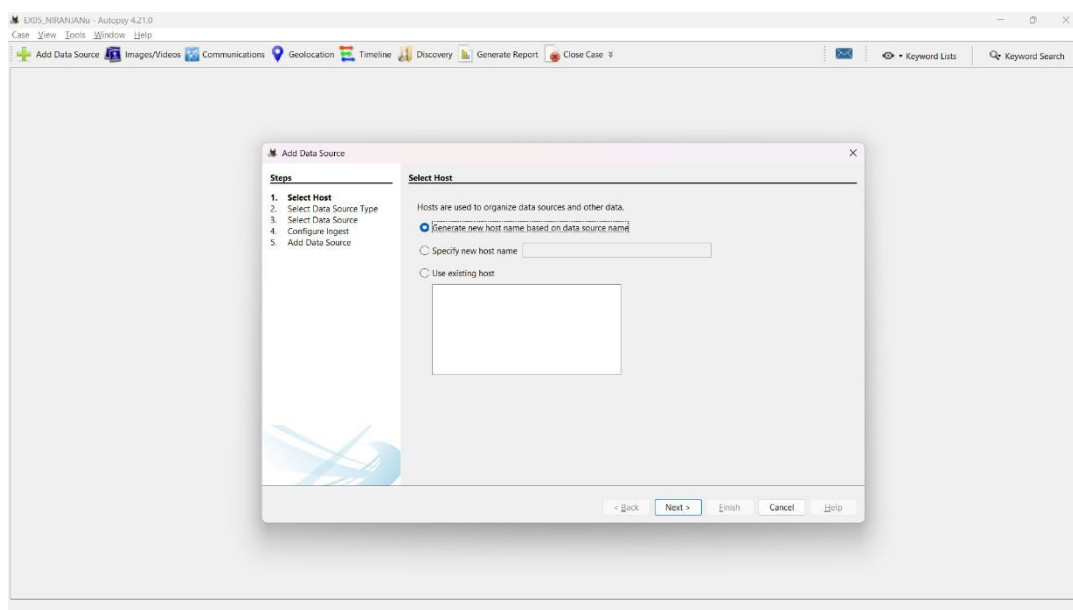


Figure 3: Selecting the data-source type.

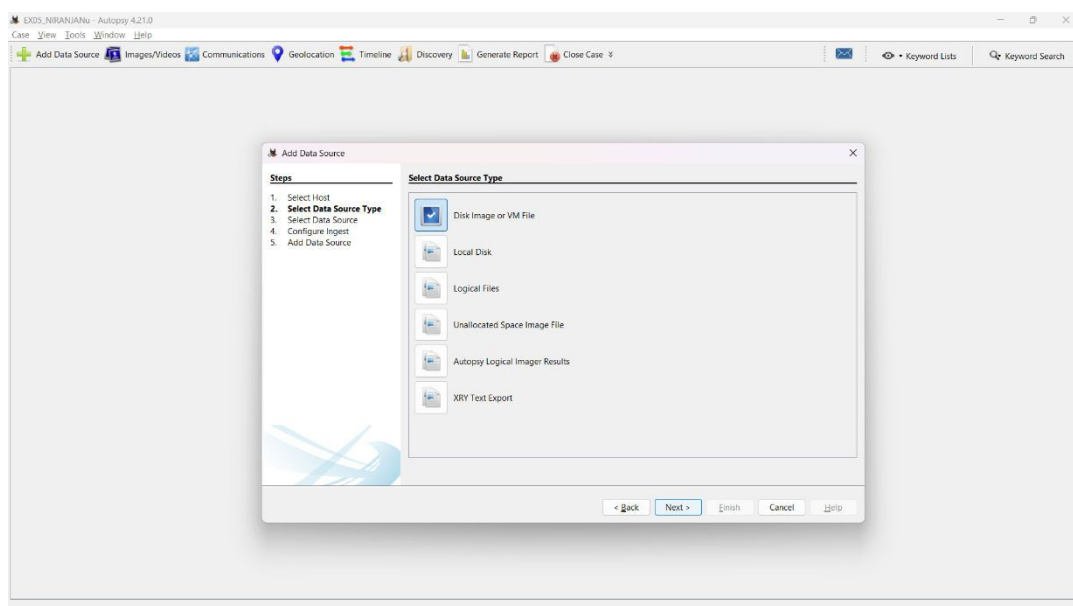


Figure 4: Selecting the E01 evidence file.

Step 3: Configure Ingest Modules

The ingest configuration screen was displayed. The selected modules included:

- Recent Activity.
- Hash Lookup.
- File Type Identification.
- Extension Mismatch Detector.
- Embedded File Extractor.
- Picture Analyzer.
- Keyword Search.
- E-mail Parser.
- Encryption Detection.
- Interesting Files Identifier.

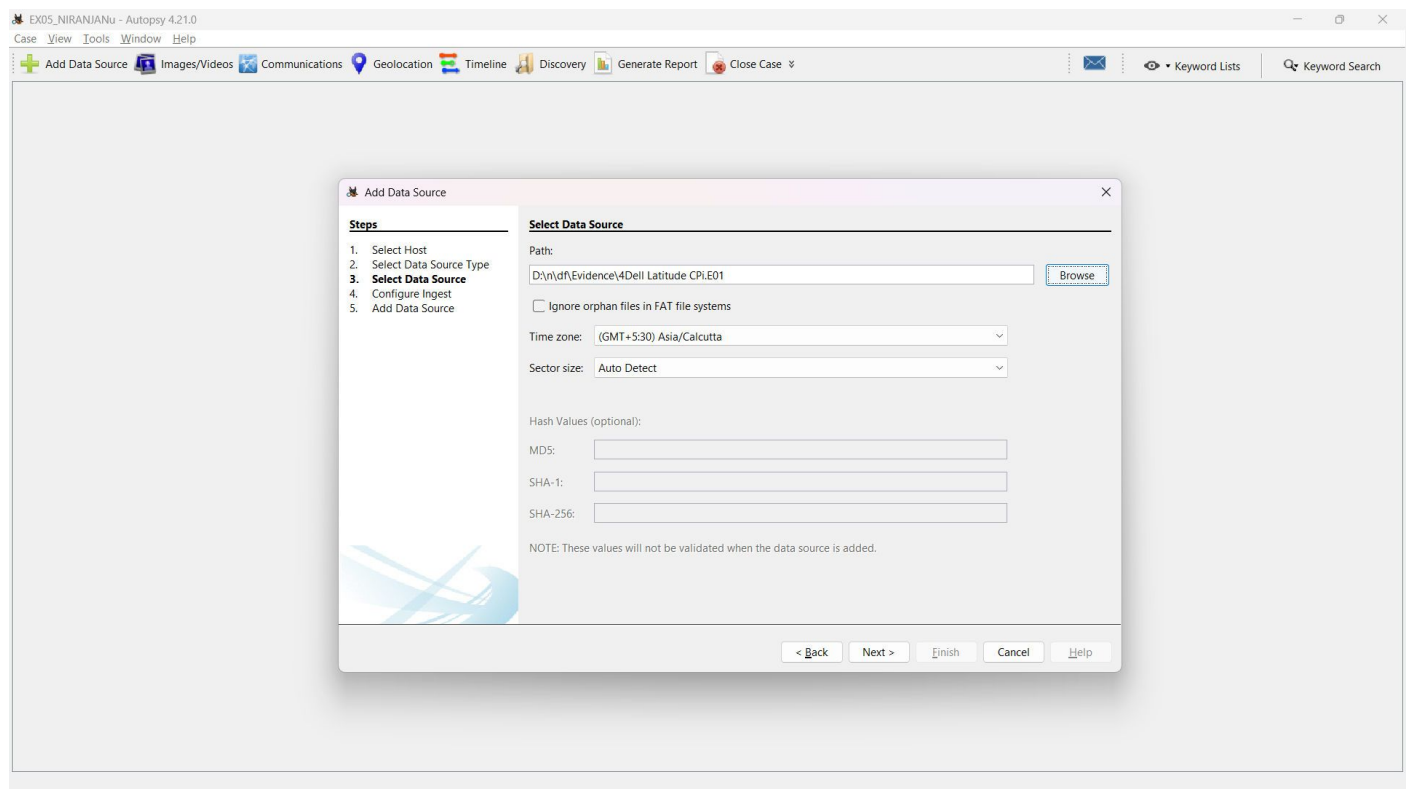


Figure 5: Configuring the Autopsy ingest modules.

Step 4: Start Evidence Processing

The data source was added to the case, and Autopsy began analyzing the image. The progress panel displayed the processing status for files, data artifacts, and analysis results.

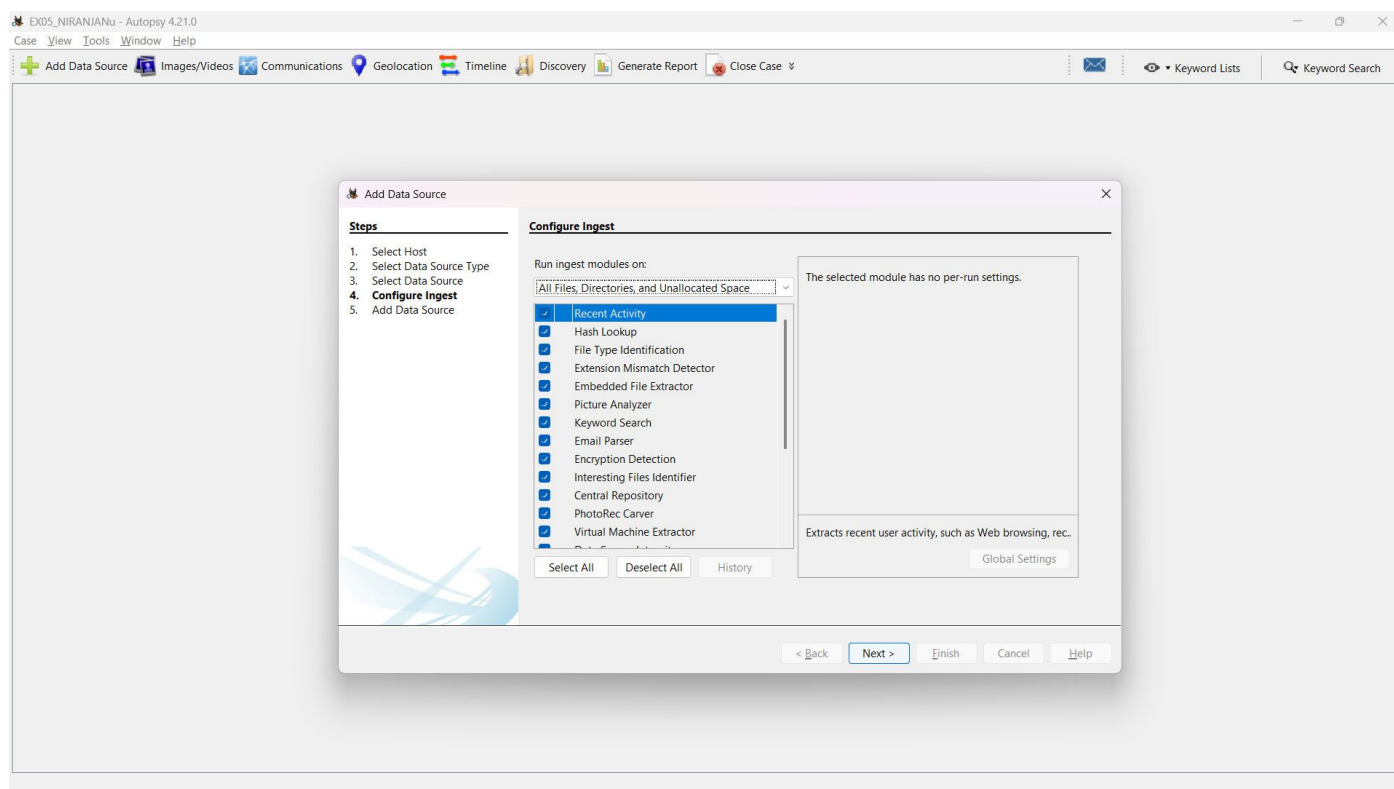


Figure 6: Data source added and analysis started.

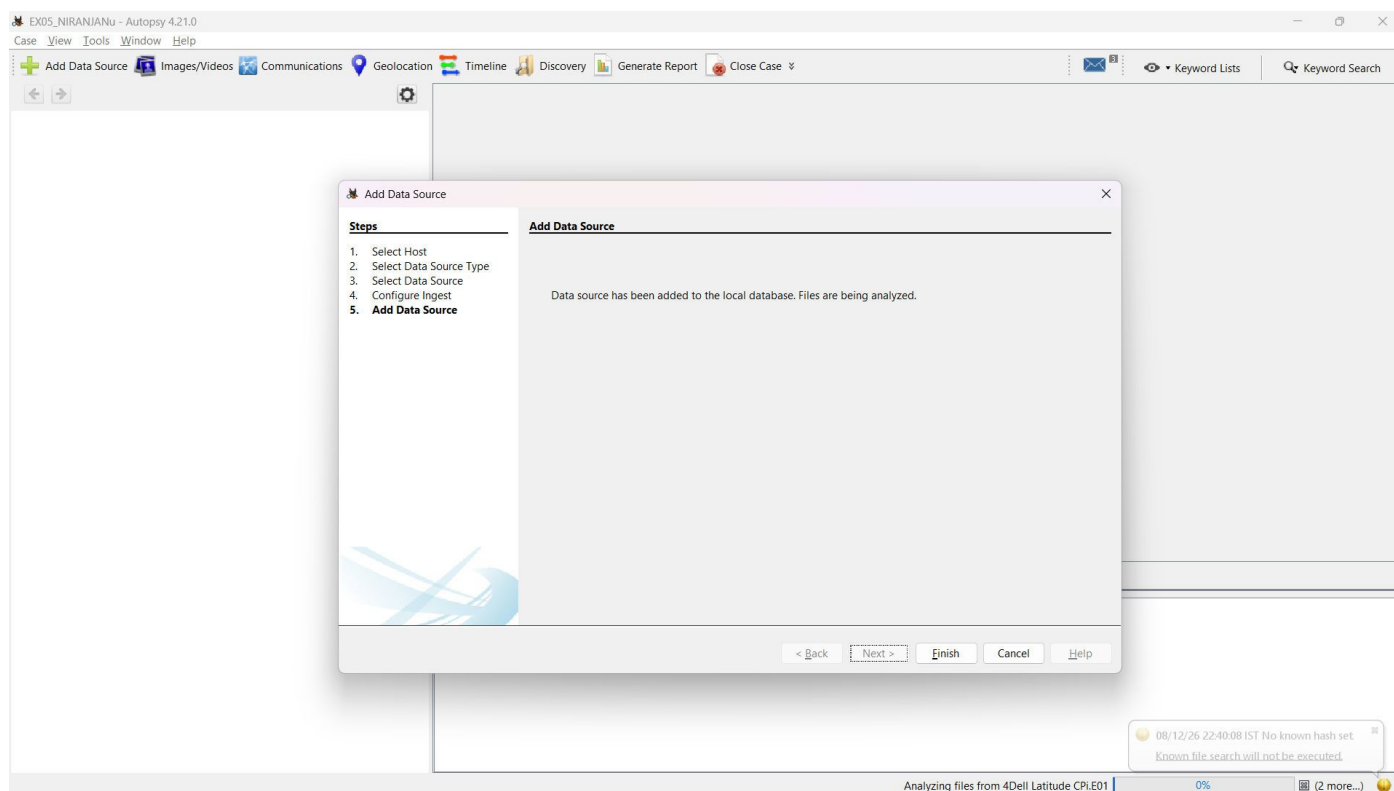


Figure 7: Ingest processing progress.

Step 5: Examine the Results

After processing began, the Autopsy tree view displayed categories such as:

- File Types.
- Deleted Files.
- File Size.
- Data Artifacts.
- Operating System Information.
- Recent Documents.
- Run Programs.
- USB Device Attached.
- Web Bookmarks.
- Web Cookies.
- Web History.
- Web Search.
- Analysis Results.

A file was selected from the image category. Autopsy displayed its metadata, including the file name, path, type, size, timestamps, allocation status, MD5 hash, SHA-256 hash, and hash-lookup result. A preview of the selected image was also displayed.

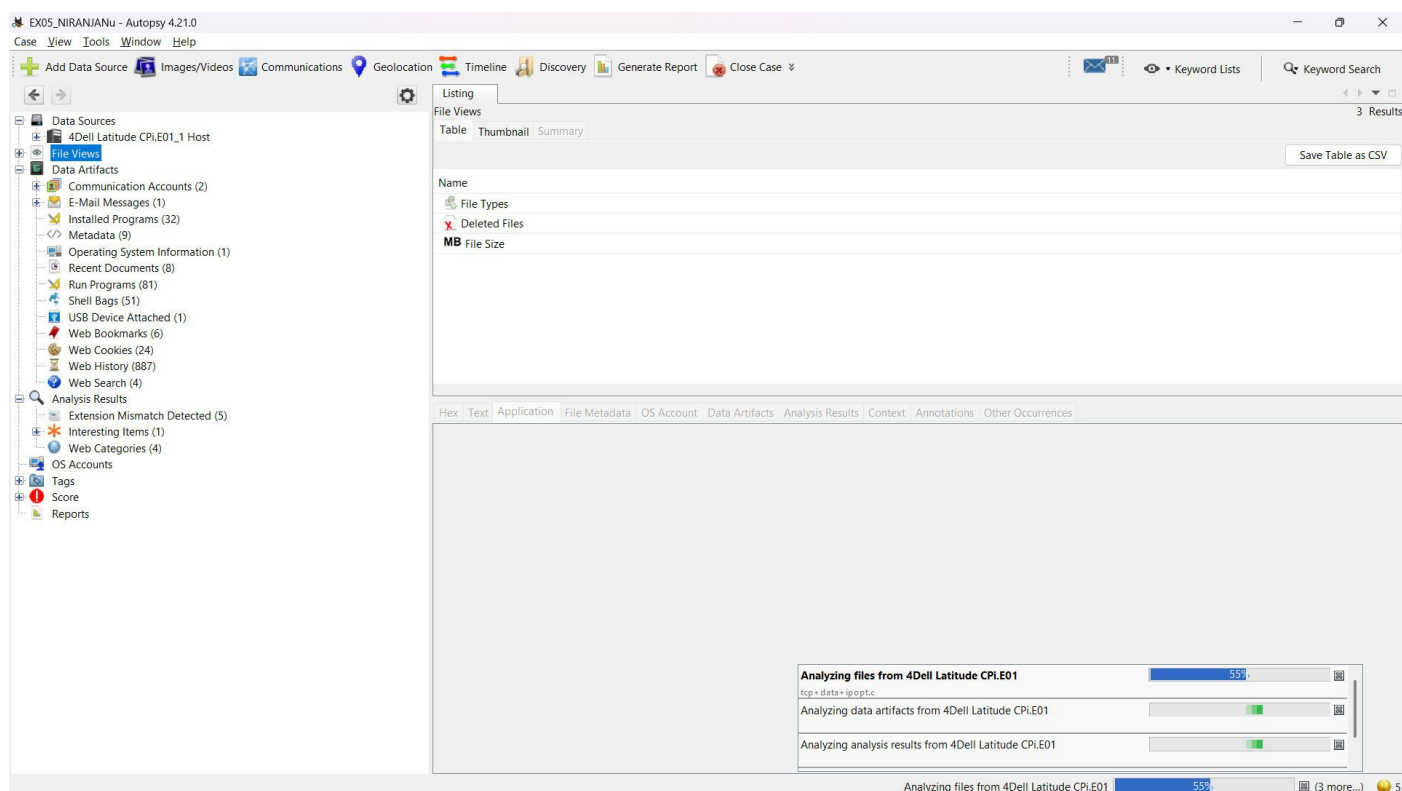


Figure 8: Autopsy File Views.

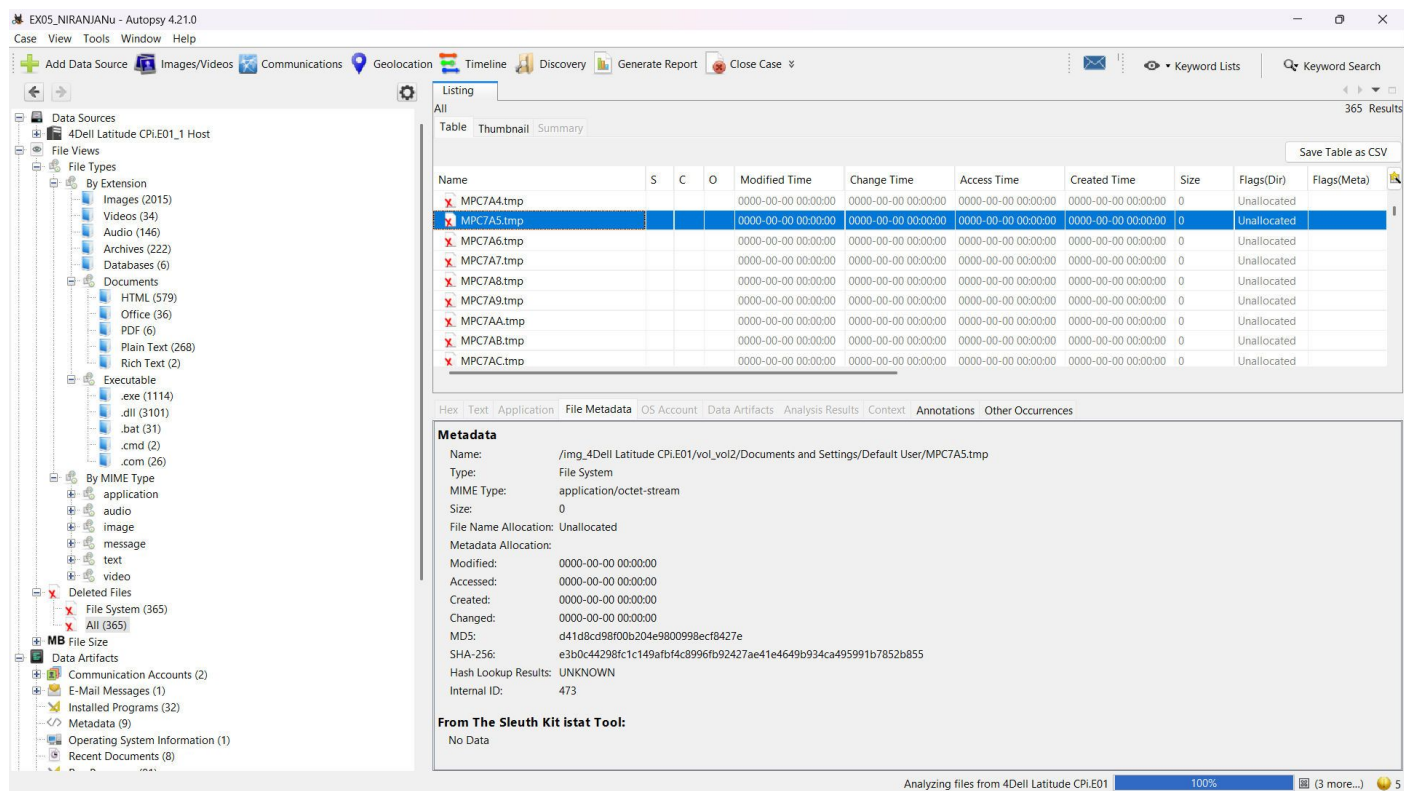


Figure 9: File metadata and preview.

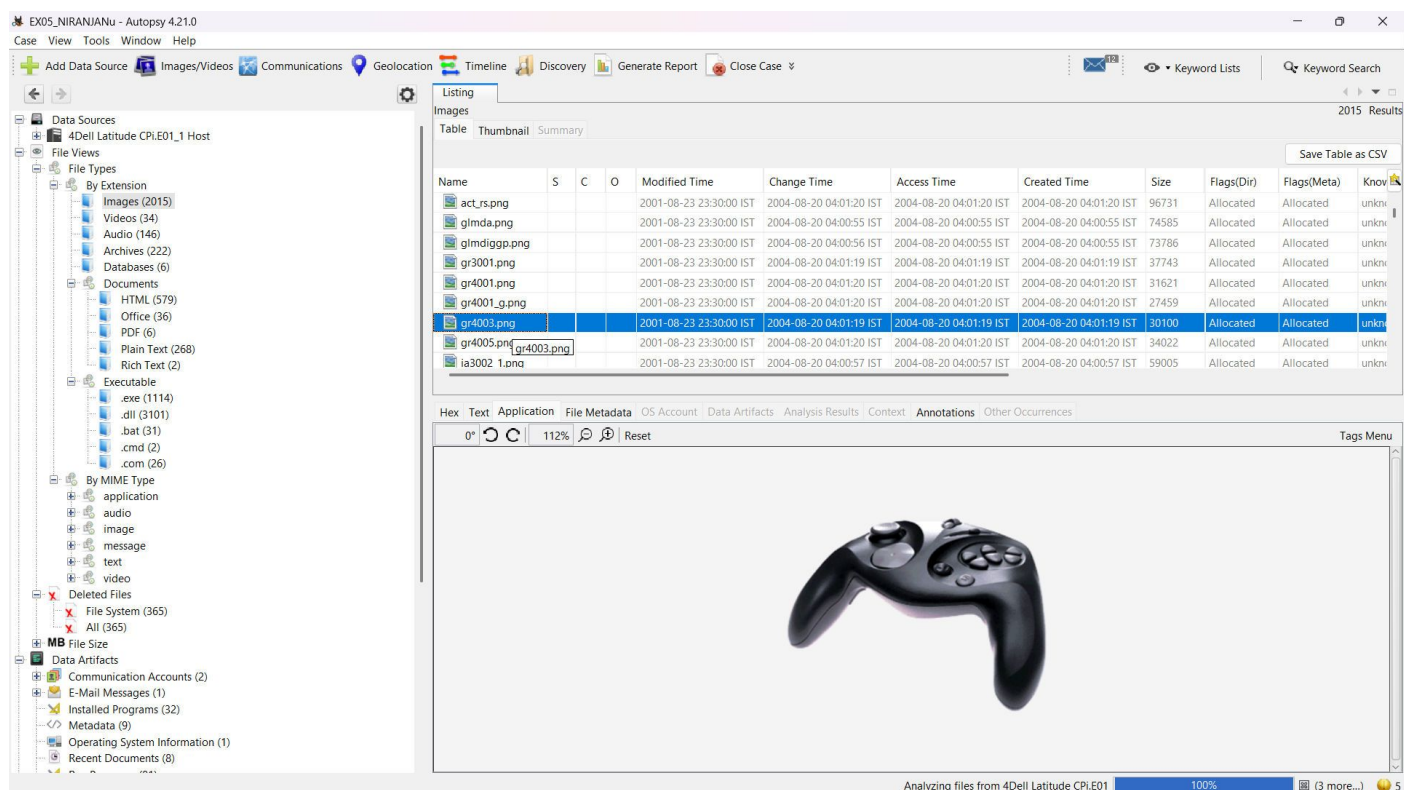


Figure 10: Image-file analysis.

EX05_NIRANJANu - Autopsy 4.21.0

Case View Tools Window Help

Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case

Keyword Lists Keyword Search

Listing

PDF Table Thumbnail Summary

6 Results

Save Table as CSV

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)	Flags(Meta)	Known	Lo
rfbproto.ndf				1998-08-28 12:42:30 IST	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	76357	Allocated	Allocated	unknown	/ir
4Dell Latitude CPiE01							0000-00-00 00:00:00	19197	Allocated	Allocated	unknown	/ir
4Dell Latitude CPiE02							0000-00-00 00:00:00	76357	Allocated	Allocated	unknown	/ir
							0000-00-00 00:00:00	19197	Allocated	Allocated	unknown	/ir
							0000-00-00 00:00:00	1202458	Allocated	Allocated	unknown	/ir
							0000-00-00 00:00:00	3536682	Allocated	Allocated	unknown	/ir

Occurrences

1/12

Save

File name: rfbprotoheader.pdf

Files of type: All Files

Save selected file

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Analyzing files from 4Dell Latitude CPiE01 100% (3 more...) 5

New tab x rfbprotoheader.pdf x +

File D:/n/df/Evidence/rfbprotoheader.pdf

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Ask Copilot

Edit with Acrobat

value in a pixel to the least significant bit. To find the red value from a given pixel, do the following:

- 1) Swap pixel value according to bigEndian (e.g. if bigEndian is false and host byte order is big endian, then swap).
- 2) Shift right by redShift.
- 3) AND with redMax (in host byte order).
- 4) You now have the red value between 0 and redMax. */

```
CARD8 greenShift; /* similar for green */
CARD8 blueShift; /* and blue */
CARD8 pad1;
CARD16 pad2;
} rfbPixelFormat;
```

rfbproto.h 3/12

```
#define sz_rfbPixelFormat 16

/*
 * Initial handshaking messages
 */

/*
 * Protocol Version
 */
```


Step 6: Generate a Report

The Generate Report option was selected. An HTML report was configured by selecting the required result types and report modules.

The following report-related screens were captured:

- Report save dialog.
- Result type selection.
- Report module selection.
- Report generation completion.
- Generated HTML/PDF report preview.

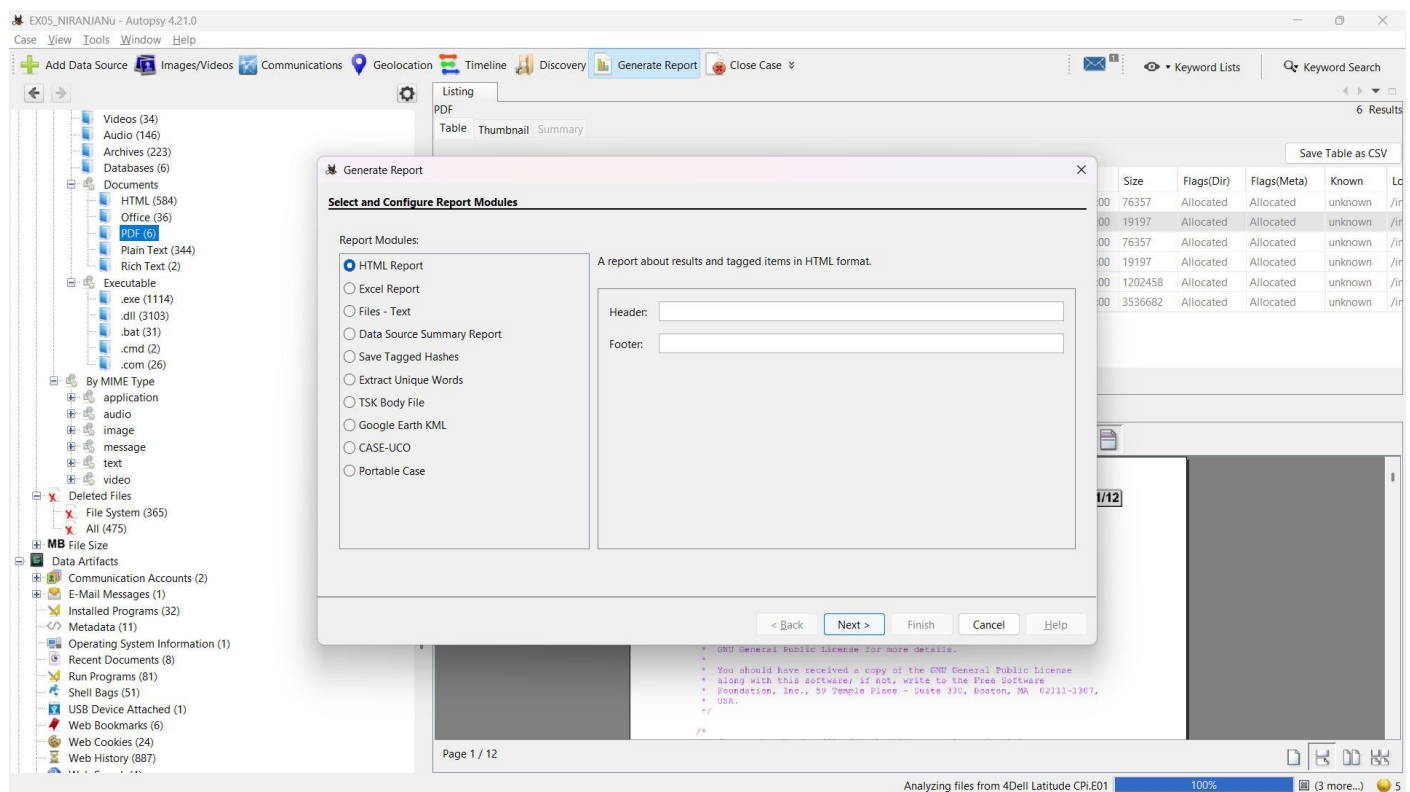


Figure 11: Report save dialog.

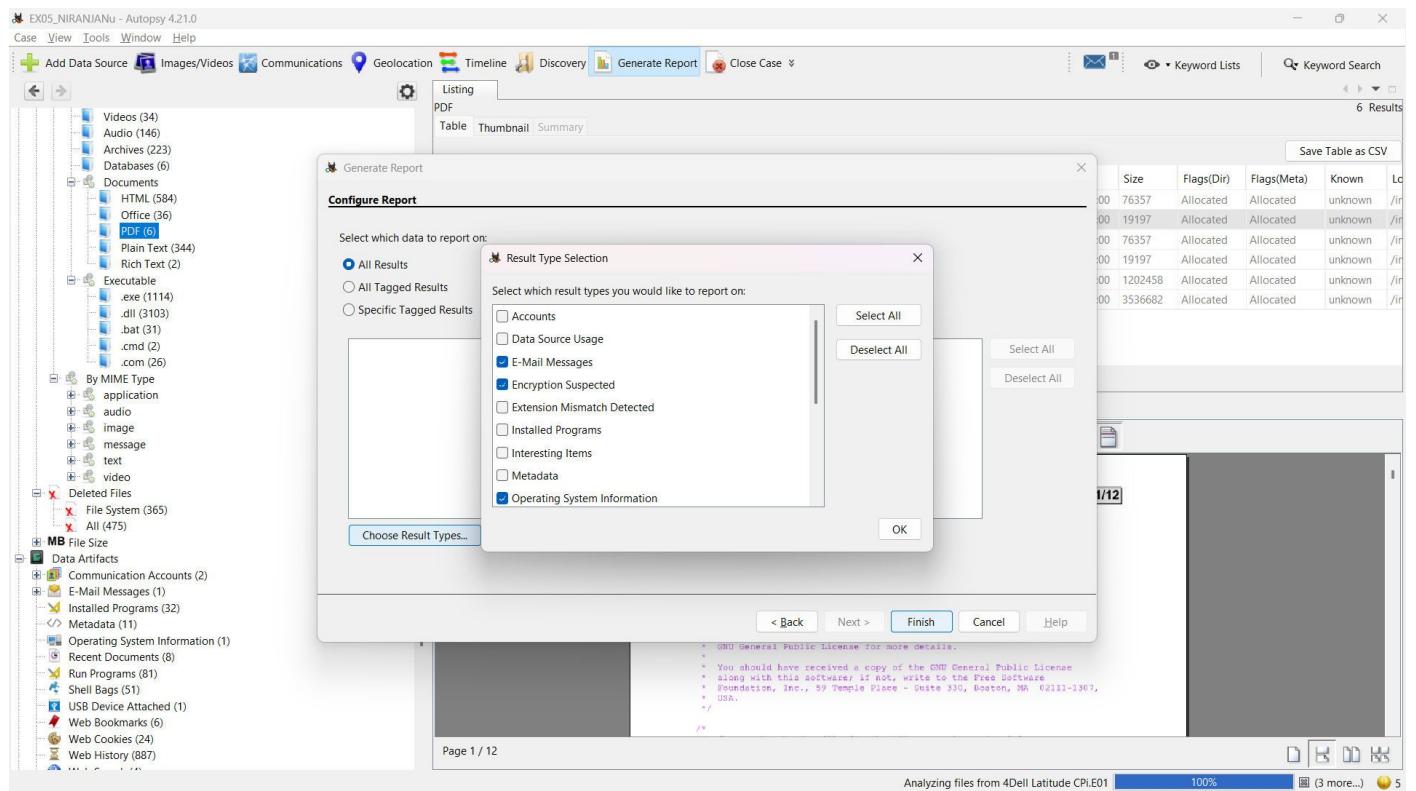


Figure 12: Report module selection.

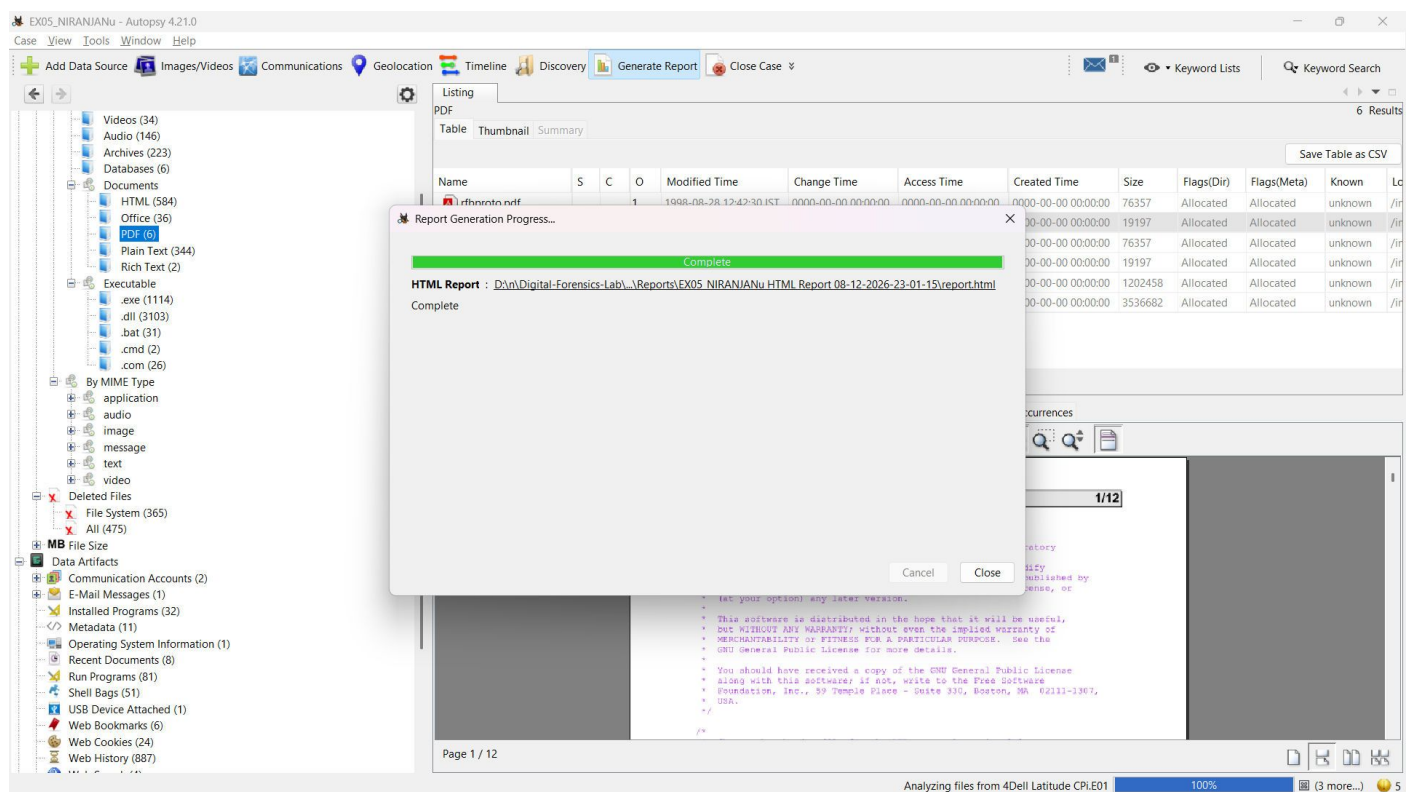


Figure 14: Report generation completed.

Autopsy Forensic Report for case

D:\n\Digital-Forensics-Lab\Ex-05-Autopsy-Case-and-Evidence-Import\EX05_NIRANJAN\Reports\EX05_NIRANJAN\30HTML%20Report%2008-12-2026-23-01-15\report.html

Autopsy Forensic Report

Warning, this report was run before ingest services completed!

HTML Report Generated on 2023/12/12 23:01:15

Report Navigation

Case Summary

E-Mail Messages (1)

Encryption Suspected (2)

Operating System Information (1)

Recent Documents (1)

Run Programs (81)

Shell Bags (51)

Tagged Files (0)

Tagged Images (0)

Tagged Results (0)

USB Device Attached (1)

Web Bookmarks (6)

Web History (887)

Web Search (1)

Case:

ex05-boomii

Case Number:

01

Number of data sources in case:

1

Examiner:

Boomii

Image Information:

4Dell Latitude CPLE01

Timezone:

Asia/Calcutta

Path:

C:/n/df/Evidence/4Dell Latitude CPi.E01

Path:

C:/n/df/Evidence/4Dell Latitude CPi.E02

Software Information:

Autopsy Version:

4.21.0

Android Analyzer Module:

4.21.0

Android Analyzer (aLEAPP) Module:

4.21.0

Central Repository Module:

4.21.0

Autopsy Forensic Report for case

D:\n\Digital-Forensics-Lab\Ex-05-Autopsy-Case-and-Evidence-Import\EX05_NIRANJAN\Reports\EX05_NIRANJAN\20HTML%20Report%2008-12-2026-23-01-15\report.html

Recent Documents

Report Navigation

Case Summary

E-Mail Messages (1)

Encryption Suspected (2)

Operating System Information (1)

Recent Documents (8)

Run Programs (81)

Shell Bags (51)

Tagged Files (0)

Tagged Images (0)

Tagged Results (0)

USB Device Attached (1)

Web Bookmarks (6)

Web History (887)

Web Search (4)

Path	Date/Time	Source File
C:\Program Files\mIRC\channels	2004-08-20 21:20:40 IST	/img_4Dell Latitude CPi.E01/vol_vol2/Documents and Settings/Mr. Evil/Recent/channels (2).lnk
C:\Program Files\mIRC\channels\channels.txt	2004-08-20 21:20:40 IST	/img_4Dell Latitude CPi.E01/vol_vol2/Documents and Settings/Mr. Evil/Recent/channels.lnk
D:\Drivers\Anonymizer	2004-08-20 20:34:52 IST	/img_4Dell Latitude CPi.E01/vol_vol2/Documents and Settings/Mr. Evil/Recent/Anonymizer.lnk
D:\Drivers\Anonymizer\keys.txt	2004-08-20 20:34:51 IST	/img_4Dell Latitude CPi.E01/vol_vol2/Documents and Settings/Mr. Evil/Recent/keys.lnk
D:\Drivers\GhostWare	2004-08-20 20:39:16 IST	/img_4Dell Latitude CPi.E01/vol_vol2/Documents and Settings/Mr. Evil/Recent/GhostWare.lnk
D:\Drivers\GhostWare\Receipt.rtf	2004-08-20 20:39:16 IST	/img_4Dell Latitude CPi.E01/vol_vol2/Documents and Settings/Mr. Evil/Recent/Receipt.lnk
No preferred path found	2004-08-26 20:38:14 IST	/img_4Dell Latitude CPi.E01/vol_vol2/Documents and Settings/Mr. Evil/Recent/Temp on m1200 (4.12.220.254).lnk
\\4.12.220.254\TEMP\yng13.bmp	2004-08-26 20:38:12 IST	/img_4Dell Latitude CPi.E01/vol_vol2/Documents and Settings/Mr. Evil/Recent/yng13.lnk

5. Observations

The following observations were made during analysis: •

The E01 evidence image was added successfully.

- The associated E02 segment was available in the evidence directory.
- Autopsy displayed file-type categories, including images, videos, audio, archives, documents, databases, and executable files.
- Deleted-file entries were displayed.
- Data artifacts included e-mail messages, communication accounts, installed programs, operating-system information, recent documents, USB-device information, bookmarks, cookies, web history, and web searches.
- A selected image file was opened in the preview pane.
- File metadata and hash values were displayed for the selected file.
- An HTML forensic report was generated.

6. Result

A single-user Autopsy case named EX05_NIRANJANu was created successfully. The 4Dell Latitude CPi.E01 evidence image was imported, and the associated 4Dell Latitude CPi.E02 segment was kept in the same evidence directory.

The evidence was processed using the selected ingest modules. Autopsy displayed file-system information, deleted files, file categories, metadata, web artifacts, and other data artifacts. A forensic HTML report was generated successfully.

7. Limitation

One generated report displayed the following warning:

text

Warning, this report was run before ingest services completed!

This means the report was generated while analysis was still running. A final report should be generated again after the ingest progress reaches completion.

8. Conclusion

Autopsy was successfully used to create a forensic case, import the split E01 evidence image, configure ingest modules, examine extracted artifacts, and generate a forensic report. The experiment demonstrated the basic workflow for analyzing a disk image in Autopsy.